

MODULE 2...

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QUES 1 :

```
FraudSystemDetection.java > FraudSystemDetection
1  import java.util.Scanner;
2
3  public class FraudSystemDetection {
4
5      Run | Debug
6      public static void main(String[] args) {
7
8          Scanner sc = new Scanner(System.in);
9
10         int n = sc.nextInt();
11
12         int[] transaction = new int[n];
13
14         for (int i = 0; i < n; i++) {
15             transaction[i] = sc.nextInt();
16         }
17
18         int search = sc.nextInt();
19
20         boolean found = false;
21
22         for (int i = 0; i < n; i++) {
23             if (transaction[i] == search) {
24                 found = true;
25                 break;
26             }
27         }
28
29         if (found)
30             System.out.println("Fraud Transaction Found");
31         else
32             System.out.println("Transaction Not Found");
33
34         sc.close();
35     }
```

OUTPUT :

```
PS C:\Users\Siddhika\Downloads\VSCode\Java> cd "c:\
tection }
5
1201 1305 1450 1408 1502
1450
Fraud Transaction Found
PS C:\Users\Siddhika\Downloads\VSCode\Java>
```

QUES 2 :

```
J ServerLogSearch.java > ...
1  import java.util.Scanner;
2
3  public class ServerLogSearch {
4
5      Run | Debug
6      public static void main(String[] args) {
7
8          Scanner sc = new Scanner(System.in);
9
10         int n = sc.nextInt();
11         int[] logs = new int[n];
12
13         for (int i = 0; i < n; i++)
14             logs[i] = sc.nextInt();
15
16         int target = sc.nextInt();
17
18         int low = 0, high = n - 1;
19
20         while (low <= high) {
21             int mid = (low + high) / 2;
22
23             if (logs[mid] == target) {
24                 System.out.println(x: "Log Found");
25                 return;
26             }
27
28             if (target < logs[mid])
29                 high = mid - 1;
30             else
31                 low = mid + 1;
32         }
33
34         System.out.println(x: "Log Not Found");
35         sc.close();
36     }
```

OUTPUT :

```
PS C:\Users\Siddhika\Downloads\VSCode\Java> cd "c:
7
1001 1005 1010 1015 1020 1030 1040
1015
Log Found
```

QUES 3 :

```
EmployeeSalarySort.java X
EmployeeSalarySort.java > EmployeeSalarySort
1  import java.util.Scanner;
2
3  public class EmployeeSalarySort {
4
5      Run | Debug
6      public static void main(String[] args) {
7
8          Scanner sc = new Scanner(System.in);
9
10         int n = sc.nextInt();
11         int[] salary = new int[n];
12
13         for (int i = 0; i < n; i++)
14             salary[i] = sc.nextInt();
15
16         // Bubble Sort
17         for (int i = 0; i < n - 1; i++) {
18             for (int j = 0; j < n - i - 1; j++) {
19                 if (salary[j] > salary[j + 1]) {
20                     int temp = salary[j];
21                     salary[j] = salary[j + 1];
22                     salary[j + 1] = temp;
23                 }
24             }
25         }
26
27         // Print sorted salaries
28         for (int i = 0; i < n; i++)
29             System.out.print(salary[i] + " ");
30
31         sc.close();
32     }
```

OUTPUT :

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Siddhika\Downloads\VSCode\Java> cd "c:\Users\Siddhika\Downloads\VSCode\Java"
ort }
5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000
PS C:\Users\Siddhika\Downloads\VSCode\Java>
```

QUES 4 :

```
EmployeeSalarySort.java WebsiteResponseSort.java X
WebsiteResponseSort.java > WebsiteResponseSort
1  import java.util.Scanner;
2
3  public class WebsiteResponseSort {
4
5      Run | Debug
6      public static void main(String[] args) {
7
8          Scanner sc = new Scanner(System.in);
9
10         int n = sc.nextInt();
11         int[] response = new int[n];
12
13         for (int i = 0; i < n; i++)
14             response[i] = sc.nextInt();
15
16         // Insertion Sort
17         for (int i = 1; i < n; i++) {
18             int key = response[i];
19             int j = i - 1;
20
21             while (j >= 0 && response[j] > key) {
22                 response[j + 1] = response[j];
23                 j--;
24             }
25             response[j + 1] = key;
26         }
27
28         // Print sorted response times
29         for (int i = 0; i < n; i++)
30             System.out.print(response[i] + " ");
31
32         sc.close();
33     }
34 }
```

OUTPUT :

```
PS C:\Users\Siddhika\Downloads\VSCode\Java> cd "c:\Users\Siddhika\Downloads\VSCode\Java" & java EmployeeSalarySort }
5
250 120 320 180 100
100 120 180 250 320
PS C:\Users\Siddhika\Downloads\VSCode\Java>
```

QUES 5 :

```
EmployeeSalarySort.java WebsiteResponseSort.java BugReportPriority.java X
BugReportPriority.java > BugReportPriority
1  import java.util.Scanner;
2
3  public class BugReportPriority {
4
5      Run | Debug
6      public static void main(String[] args) {
7
8          Scanner sc = new Scanner(System.in);
9
10         int n = sc.nextInt();
11         int[] priority = new int[n];
12
13         for (int i = 0; i < n; i++)
14             priority[i] = sc.nextInt();
15
16         // Selection Sort
17         for (int i = 0; i < n - 1; i++) {
18             int min = i;
19
20             for (int j = i + 1; j < n; j++) {
21                 if (priority[j] < priority[min])
22                     min = j;
23             }
24
25             int temp = priority[i];
26             priority[i] = priority[min];
27             priority[min] = temp;
28         }
29
30         // Print sorted priorities
31         for (int i = 0; i < n; i++)
32             System.out.print(priority[i] + " ");
33
34         sc.close();
35     }
```

OUTPUT

```
PS C:\Users\Siddhika\Downloads\VSCode\Java> cd :
y }
5
9 2 8 1 5
1 2 5 8 9
PS C:\Users\Siddhika\Downloads\VSCode\Java>
```


Date ___/___/___

Day _____

Module 2

Q.1)

```
import java.util.Scanner;
```

```
public class FraudSystemDetection {
```

```
    public static void main (String [] args) {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        int n = sc.nextInt();
```

```
        int [] transaction = new int [n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            transaction[i] = sc.nextInt();
```

```
        }
```

```
        int search = sc.nextInt();
```

```
        boolean found = false;
```

```
        for (int i = 0; i < n; i++) {
```

```
            if (transaction[i] == search) {
```

```
                found = true;
```

```
            } break;
```

if (found) {

system.out.println(x: "Fraud Transaction found");

else

system.out.println(x: "Transaction not found");

sc.close();

}

OUTPUT

→ 5

→ 1201 1305 1480 1408 1502

→ 1450.

Fraud Transaction found.

Date ____/____/____

Day _____

Q2)

```
import java.util.Scanner ;

public class ServerLogSearch {

    public static void main (String [] args) {

        Scanner sc = new Scanner (System.in);
        int n = 5 int[] logs = new int[n];

        for (int i = 0; i < n; i++)
            log[i] = sc.nextInt();

        int target = sc.nextInt();

        int low = 0, high = n - 1;

        while (low <= high) {
            int mid = (low + high) / 2;

            if (logs[mid] == target) {
                System.out.println ("Log Found");
                return;
            }
        }
```

```
if (target < logs[mid])
```

```
    high = mid - 1;
```

```
else
```

```
    low = mid + 1;
```

```
}
```

```
System.out.println(x: "Log not found");
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT

→ 7

→ 1001 1005 1010 1015 1020 1030 1040

→ 1015

⇒⇒ Log found.

Date ___/___/___

Day _____

Q3.)

```
import java.util.Scanner;
```

```
public class EmployeeSalarySort {
```

```
    public static void main (String [] args) {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        int n = sc.nextInt();
```

```
        int [] salary = new int [n];
```

```
        for (int i = 0; i < n; i++)
```

```
            salary[i] = sc.nextInt();
```

```
        for (int i = 0; i < n-1; i++) {
```

```
            for (int j = 0; j < n-i-1; j++) {
```

```
                if (salary[j] > salary[j+1]) {
```

```
                    int temp = salary[j];
```

```
                    salary[j] = salary[j+1];
```

```
                    salary[j+1] = temp;
```

```
                }
```

```
            }
```

```
for (int i = 0; i < n; i++)  
    System.out.print(salary[i] + " ");  
sc.close();  
}
```

OUTPUT

⇒ 5

⇒ 45000 32000 55000 28000 60000

⇒ 28000 32000 48000 55000 60000

Date ___/___/___

Day _____

Q4)

```
import java.util.Scanner;
```

```
public class websiteResponseSort {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        int[] response = new int[n];
```

```
        for (int i = 0; i < n; i++)
```

```
            response[i] = sc.nextInt();
```

```
        for (int i = 1; i < n; i++) {
```

```
            int key = response[i];
```

```
            int j = i - 1;
```

```
            while (j >= 0 && response[j] > key)
```

```
                response[j+1] = response[j];
```

```
                j--;
```

```
        }
```




```
    response[j+1] = key;  
}
```

```
for (int i = 0; i < n; i++)
```

```
    System.out.print(response[i] + " ");
```

```
sc.close();
```

```
}
```

OUTPUT.

⇒ 5

⇒ 250 120 320 180 100

⇒ 100 120 180 250 320

Date ___/___/___

Day _____

Q5)
import

java.util.Scanner;

public class BugReportPriority {

public static void main (String[] args) {

Scanner sc = new Scanner (System.in);

int n = sc.nextInt();

int [] priority = new int [n];

for (int i = 0; i < n; i++)

priority [i] = sc.nextInt();

for (int i = 0; i < n - 1; i++) {

int min = i;

for (int j = i + 1; j < n; j++) {

if (priority [j] < priority [min])

min = j;

}

→

```
int temp = priority[i];  
priority[i] = priority[min];  
priority[min] = temp;  
}
```

```
for (int i = 0; i < n; i++)  
    system.out.print(priority[i] + " ");  
sc.close();
```

```
}
```

OUTPUT

→ 5

⇒ 9 2 8 1 5

⇒ 12 5 8 9